CONFIDENTIAL

FACULTY OF COMPUTING
UTM Johor Bahru

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER 1 2023/2024

SUBJECT CODE	**: SECP 3213**
SUBJECT NAME	: **BUSINESS INTELLIGENCE**
YEAR/COURSE	**: 3/SECPH**
TIME	**: 3 HOURS**
DATE/DAY	**:**
VENUE	**:**

INSTRUCTIONS TO THE STUDENTS:

1. For Section A, write the answer on answer sheets.
2. For Section B and C, saved the file in the allocated folder at your PC.
3. A candidate who is suspected of cheating in examination is liable to disciplinary action including (but not limited to) suspension or expulsion from the university. All materials and/or devices which are found in violation of any examination rules and regulation will be confiscated.

NAME	
MATRIC NO.	
SECTION	
LECTURER	

(This question paper consists of 5 pages including this page)

SECTION A **[10 MARKS]**

Circle the correct answer.

1. What is the primary purpose of a data warehouse in the context of Business Intelligence?

    a) Storing only raw, unprocessed data.
    b) Serving as a backup for all organizational data.
    c) Providing a centralized repository for structured and organized data.
    d) Managing real-time data.

2. Which type of Business Intelligence analytics focuses on suggesting the best course of action?

    a) Descriptive analytics.
    b) Predictive analytics.
    c) Prescriptive analytics.
    d) Diagnostic analytics.

3. What is the role of a key performance indicator (KPI) in Business Intelligence?

    a) Measuring employee satisfaction.
    b) Assessing environmental impact.
    c) Providing a quantifiable metric for evaluating performance against objectives.
    d) Analyzing market trends.

4. In the context of Business Intelligence reporting, what does OLAP stand for?

    a) Online Learning and Assessment Platform.
    b) Online Analytical Processing.
    c) Offline Analysis and Planning.
    d) Overhead Labor Allocation Process.

5. What is the significance of data governance in Business Intelligence?

    a) Ensuring all employees have access to all data.
    b) Safeguarding data quality, security, and compliance.
    c) Deleting historical data for simplicity.
    d) Allowing unrestricted data sharing across department.

6. Which Business Intelligence technique involves analyzing textual data to uncover patterns and insights?

    a) Sentiment analysis.
    b) Social media monitoring.
    c) Data visualization.
    d) Predictive modeling.

7. What is the primary function of a BI query tool?

    a) Generating random data for analysis.
    b) Extracting, transforming, and loading data.
    c) Allowing users to request specific information from the database.
    d) Providing real-time data updates.

8. How does Business Intelligence contribute to competitive advantage?

    a) By reducing data accessibility for competitors.
    b) By relying solely on internal expertise.
    c) By providing actionable insights that competitors lack.
    d) By keeping all data confidential.

9. What is the purpose of a DataMart in Business Intelligence?

    a) Storing only raw, unprocessed data.
    b) Providing a subset of data specific to a particular business function or team.
    c) Managing external data sources only.
    d) Restricting access to data for security purposes.

10. In Business Intelligence, what role does data granularity play in analysis?

    a) It refers to the size of the data storage facility.
    b) It determines the physical location of the data center.
    c) It defines the level of detail captured in the data.
    d) It specifies the duration for which data is stored.

SECTION B **[50 MARKS]**

For this section, you need to design a workflow using Alteryx Designer based on the following task:

1. Read all datasets below using one input data tool.
    - *ProductOrders_2010.xlsx*
    - *ProductOrders_2011.xlsx*
    - *ProductOrders_2012.xlsx*
    - *ProductOrders_2013.xlsx*

    (8 marks)

2. Add another input data tool to read dataset: *Product_Returned.xlsx*

    (5 marks)

3. Join the dataset of product orders with the dataset of product returned.

    (15 marks)

4. Union the dataset of product orders that do not return with product orders that has been returned. Ignore the datasets from R-anchor in Join tool.

    (10 marks)

5. Update the field/column **STATUS** that null with statement "Not Return".

    (10 marks)

6. Save final output as *ProductOrders20102013.hyper*

    (2 marks)

SECTION C **(40 MARKS)**

For this section, you need to create a workbook in Tableau Desktop. Load the dataset that has been prepared in Section B (i.e., *ProductOrders20102013.hyper*) into your new workbook. **Solution for each question must be visualized at different worksheet and rename it based on question no**. (Example: Q1)

1. Find the percentage of sales for items that shipped using a "Medium Box" at the following date:
    - January 2010
    - January 2011
    - January 2012
    - January 2013

    (6 marks)
2. Which state has lost profit in two consecutive years, 2012-2013? (4 marks)
3. State the region, product category and product sub-category that has lost profit (i.e., negative profit). (5 marks)
4. Determine the average, upper bound and lower bound with 95% Confidence Interval (CI) of sales value from 2010 to 2013. (5 marks)
5. Estimate (forecast) the sales value for year 2014. (5 marks)
6. Draw the scatter plot to identify which product (by Product Category > Product Name) that has highest shipping cost but low sales value. (3 marks)
7. Show the product (by Product Name) that were returned more than one. (2 marks)
8. How many orders in East Region that has been returned? (2 marks)
9. Which region has highest *SUM(Shipping cost)* for returned orders? Show the region and the shipping cost value. (3 marks)
10. How many customers in 2012 are the same person that make purchased in 2010? (5 marks)

End of Questions